

GROUP 7A

PRESENTING ON THE AGEING EFFECT ON THE KIDNEY

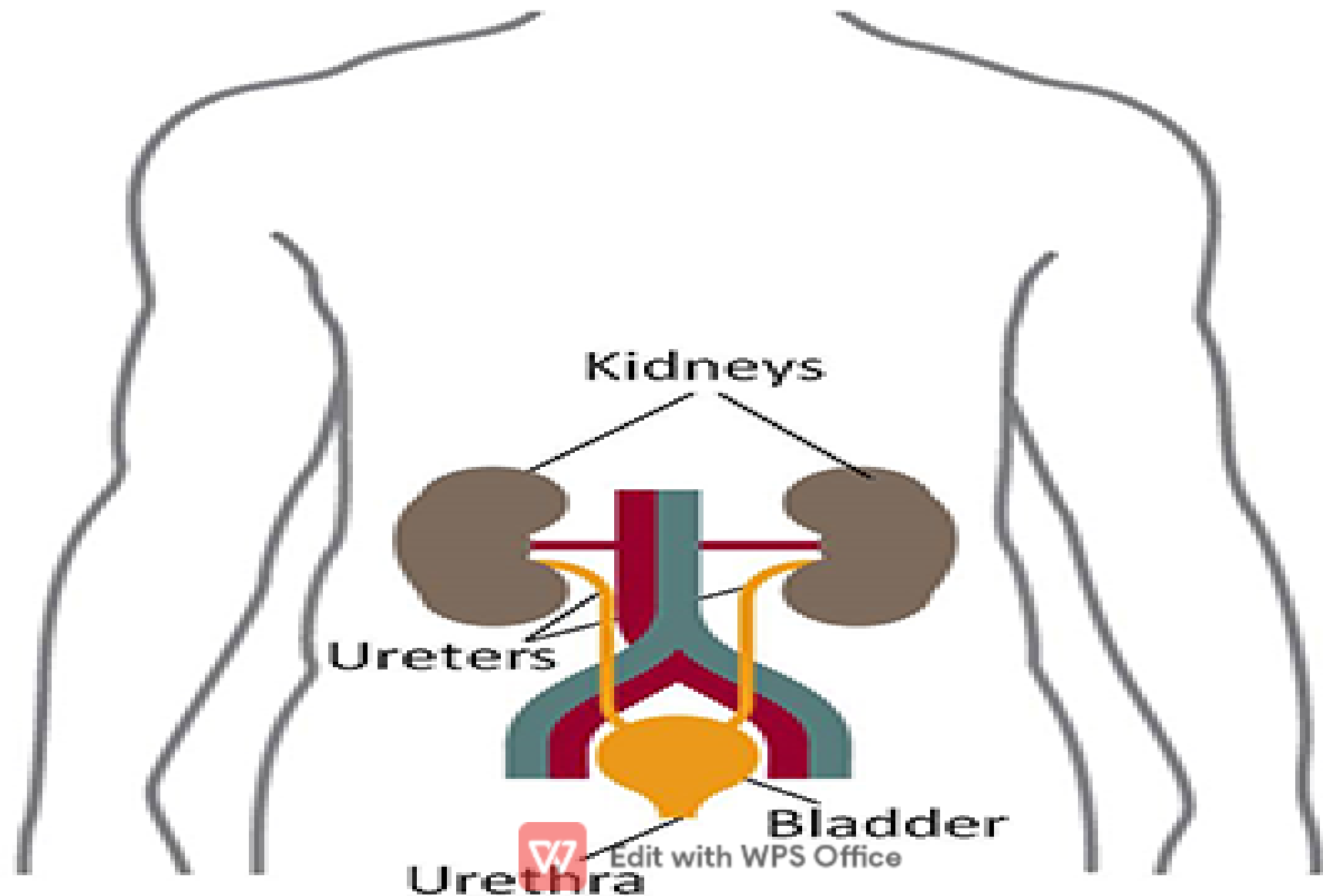


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THE KIDNEYS

- The kidneys are two bean shaped organs , each about the size of a fist
- They are located just below the rib cage , one on each side of your spine
- Healthy kidneys filter about a half cup of blood every minute , removing waste and water to make urine .
- The urine flows from the kidneys to the bladder through two thin tubes of muscle called ureters one on each side of your bladder .
- Your kidneys , ureters and bladder are part of your urinary tract .





FUNCTION OF THE KIDNEYS

It filters the blood and help remove waste and extra fluid from the body such as urea , creatinine and uric acid.

The kidneys maintain a healthy balance of water , salts ,and minerals

The kidneys also make hormones that help;

- ✓ Control your blood pressure**
- ✓ Make red blood cells**
- ✓ Keep your bones strong and healthy**



EFFECT OF AGEING ON THE KIDNEYS

With ageing the kidney undergoes structural and functional decrease in change.

STRUCTURAL CHANGES

Decrease renal mass

Renal cortex

Increased glomeruli sclerosis



FUNCTIONAL CHANGES

- i. **Decrease in the ability of the kidney to eliminate urea, creatinine, uric acid and toxins**
- ii. **Only one third of the kidney is required to maintain homeostasis**

But in every 10 years there is a decrease in the structure and function of the kidneys by 10% from the age of 20 years to 50 years.



NEPHRON

As one ages the number of nephron decreases

GLOMERULI

By the age of 80 years 40% of it will lost it total function. About 30% will decrease will not have lumen to allow blood flow.

The filtration rate decreases due to reduction of in renal blood flow(from decreases cardiac output).



MEDULLA

Glomerulosclerosis is accompanied by the formation of a direct channel between afferent and efferent arterioles, maintaining the medullary blood flow and potentially by passing the glomerulus leading to ischemia and reduction of function.



THE KIDNEY IN TOTAL

The size of the kidney decrease from reduced renal tissue growth.

The amount of extracellular fluid increases while cell mass increases.

The arteries supply in the kidney are narrowed, both kidney needs to function at their full capacity to carry out all the normal kidney function. With respect to the above, even a minor damage to one or both kidneys may result in the loss of kidney function



REFERENCE

Tabloski, P . A, Gerontological nursing, 2ND edition

<http://en.m.Wikipedia.org/wiki/kidney#effectonageing>

www.medlineplus.gov

THANK YOU



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